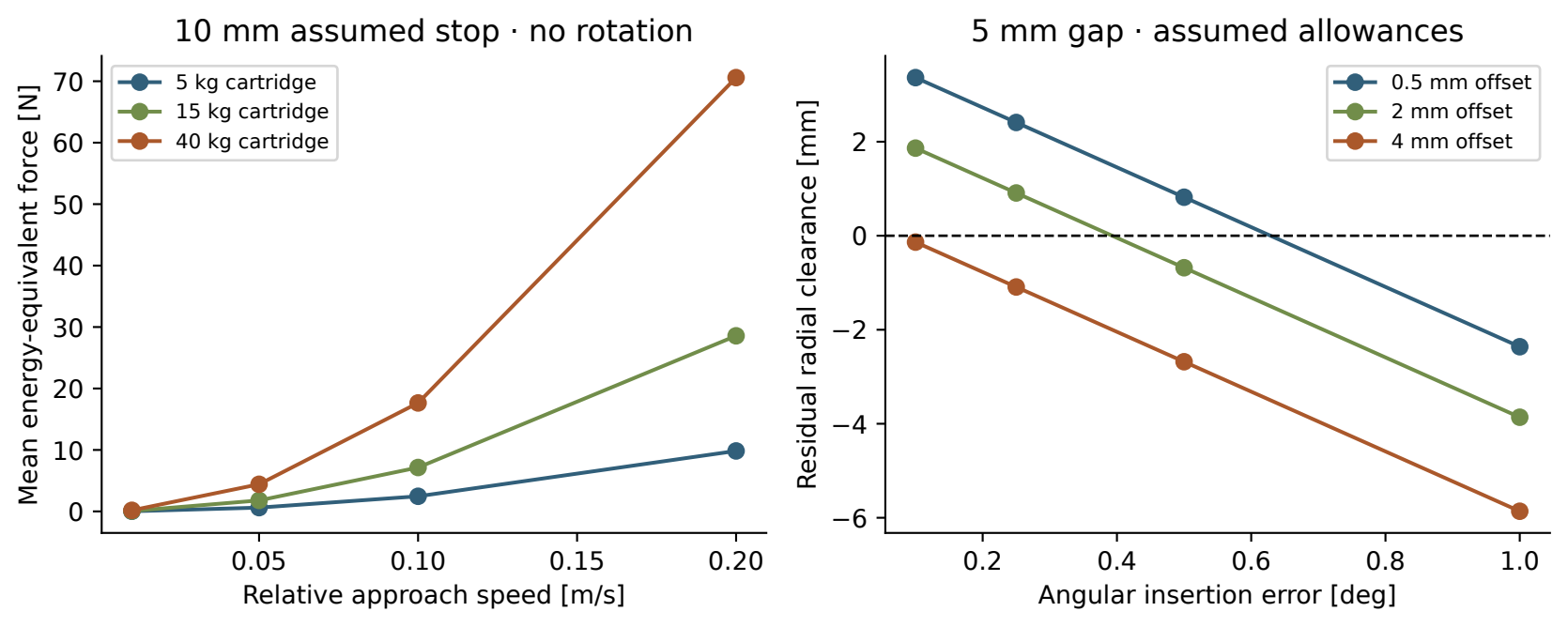
10 mm assumed stop · no rotation
5 kg cartridge
15 kg cartridge
40 kg cartridge
Mean energy-equivalent force [N]
0
10
20
30
40
50
60
70
0.05
0.10
0.15
0.20
Relative approach speed [m/s]
5 mm gap · assumed allowances
0.5 mm offset
2 mm offset
4 mm offset
Residual radial clearance [mm]
2
0
−2
−4
−6
0.2
0.4
0.6
0.8
1.0
Angular insertion error [deg]